

1. DEFINITION

DEFINITION	
Mixture of : Lauroyl macrogol-32 glycerides EP / Lauroyl polyoxyl-32 glycerides NF / Lauroyl macrogolglycerides (32) ChP and Macrogols 6000 EP / Polyethylene Glycol 6000 NF	

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Lauroyl Polyoxyl-32 Glycerides (and) PEG-150
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	H5ZC52369M (and) 30IQX730WE
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS and COMPOSITION	
Simple physical mixture of 2 components:	
Full Labeling (Key letters system) $A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Lauroyl Polyoxyl-32 Glycerides (and) Polyethylene glycol 6000	$50.0\% \leq A_2 < 75.0\%$ $25.0\% \leq B < 50.0\%$
Other constituents (i.e. incidental ingredients)	
BHT < 40 ppm (coming from the raw material used)	



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4. DRUG MASTER FILE

A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs. The submission of a DMF is not required by law or FDA regulation. A DMF is submitted solely at the discretion of the holder. (Source: US FDA)

FDA usually does not review DMFs for compendial excipients unless there is a specific reason to do so (e.g., new route of administration or application for existing excipients).

(Source: IPEC Americas)

We have no DMF for Gelucire 59/14 because it is a mixture of compendial excipients.

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopoeia Current edition	Lauroyl macrogol-32 glycerides Macrogols 6000
USP/ NF Current edition	Lauroyl polyoxyl-32 glycerides Polyethylene Glycol 6000
Japanese Pharmaceutical Excipient JPE Current edition	/
Chinese Pharmacopoeia Current edition	Lauroyl macrogolglycerides (32) /

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓(5) (USFA 172.736) (USFA 172.820)	✓(8)	✓(8) (40CFR180.910)



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EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Lauroyl macrogolglycerides or Lauroyl polyoxyglycerides	Not available as an Active Ingredient in any application. Available for use as an Excipient Ingredient in: Biologicals, Export Only, Prescription Medicines. Not available as an Equivalent Ingredient in any application.	Not authorized in listed medicines.
	Macrogol 6000	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines	Not available as an Equivalent Ingredient in any application
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			



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8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (OECD 404)	Dermal	Rabbit	GELUCIRE® 44/14 No dilution	Non Irritant Primary irritation index: 0.2	(Gattefossé data, GAT-03631, 2003)
Acute irritation (OECD 405)	Ocular	Rabbit	GELUCIRE® 44/14 No dilution	Slightly Irritant Ocular irritation index (acute): 7.3 Ocular irritation index (48h): 0.7	(Gattefossé data, GAT-03632, 2003)
Acute toxicity (OECD 401)	Oral	Rat	GELUCIRE® 44/14 pure	LD50 _(oral) > 2004 mg/kg	(Gattefossé data, GAT-97334, 1997)
Subchronic 7-day toxicity study (DRf)	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	MTD _(oral) > 2400 mg/kg/day NOEL _(oral) : 2400 mg/kg/day	(Gattefossé data, GAT- WIL261007, 2003)
Subchronic 28-day toxicity study	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	NOEL _(oral) > 2400 mg/kg/day	(Gattefossé data, GAT- WIL261008, 2003)
Subchronic 13-week toxicity study	Oral (Capsule)	Dog	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 400, 1000 and 2500 mg/kg/day	NOAEL _(oral) > 2500 mg/kg/day	(Gattefossé data, GAT- WIL 261002, 1995)
Reprotoxicity study Embryo-fetal development (Segment II)	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	NOAEL _{Matern} : 1500 mg/kg/day NOAEL _{Embryo/fetal} : 2400 mg/kg/day	(Gattefossé data, GAT- WIL 261011, 2004)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	GELUCIRE® 44/14	Negative. Not mutagenic.	(Gattefossé data, GAT- Pharmak PH301-GAF-001-95, 1996)
Mouse Lymphoma Assay	In vitro	Mouse lymphoma cell	GELUCIRE® 44/14	Negative.	Gattefossé data, GAT-Pharmak PH313-GAF-001-95, 1996)
Mammalian Erythrocyte Micronucleus Test	IP	Mouse	GELUCIRE® 44/14	Negative. Not clastogenic.	(Gattefossé data, Pharmak PH309- GAF-001-95, 1996)

OECD : Organisation for Economic Co-operation and development - JORF: Journal officiel République Française (Official Journal French Republic)

Summary of files available upon request, please contact your Gattefossé local representative.
Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview"
for a complete review on regulatory, tox and safety aspects.



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9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	EC #	Chemical name
Lauroyl Polyoxyl-32 Glycerides	57107-95-6 (or 9004-81-3 + 27638-00-2)	Not applicable (or Not applicable + 248-586-9)	Poly(oxy-1,2-ethanediyl), α,α',α'' -1,2,3-propanetriyltris[ω -hydroxy-, dodecanoate (or) Poly(oxy-1,2-ethanediyl), α -(1-oxododecyl)- ω -hydroxy- + Dodecanoic acid, diester with 1,2,3-propanetriol)
PEG 6000	25322-68-3	500-038-2	Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxy-

EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list.
EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5.
Substances without EC numbers were assigned "list numbers" in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDL)	Australia (AIC)	China (IECSC)	Other inventories
Lauroyl Polyoxyl-32 Glycerides	Listed	Listed NDSL	Listed	Listed	Poly(oxy-1,2-ethanediyl), α,α',α'' -1,2,3-propanetriyltris[ω -hydroxy-, dodecanoate: IECSC, NDSL, REACH, TCSI, TSCA, VNECI / Poly(oxy-1,2-ethanediyl), α,α',α'' -1,2,3-propanetriyltris[ω -hydroxy-, dodecanoate: AIC (or) Poly(oxy-1,2-ethanediyl), α -(1-oxododecyl)- ω -hydroxy-: AIC, DSL, ENCS, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI / Poly(oxy-1,2-ethanediyl), α -(dodecanoyl)- ω -hydroxy-: IECSC / α -(1-Oxododecyl)- ω -hydroxypoly(oxy-1,2-ethanediyl): AREC, ECL + Dodecanoic acid, diester with 1,2,3-propanetriol: AIC, DSL, ENCS, IECSC, NZIoC, PICCS, TCSI, TSCA / lauric acid, diester with glycerol: EINECS, REACH / Dodecanoic acid diester with 1,2,3-propanetriol: AREC, ECL / Diester of dodecanoic acid and 1,2,3-propanetriol: VNECI
PEG 6000	Listed	Listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxy-: AIC, DSL, ENCS, IECSC, NLP, NZIoC, PICCS, TCSI, TSCA / 3,6,9,12,15,18,21,24,27,30,33,36,39-tridecaoxahentetracontane-1,41-diol:



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NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
					EINECS, REACH, VNECI / Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxy-Ethane-1,2-diol, ethoxylated: REACH / Polyethylene Glycol: PICCS, REACH / Ethane-1,2-diol, ethoxylated: NLP / α -HYDRO- ω -HYDROXYPOLY(OXY-1,2-ETHANEDIYL): AREC, PICCS / Macrogol: PICCS, VNECI
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Lauroyl Polyoxyl-32 Glycerides (and) PEG 6000	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



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10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
LAUROYL PEG-32 GLYCERIDES (UNII: H5ZC52369M)	ORAL	CAPSULE		7200 mg
	ORAL	TABLET	0.15mg	
LAUROYL POLYOXYLGLYCERIDES (UNII: NA)	ORAL	CAPSULE	218 mg	
	ORAL	TABLET, FILM COATED	0.15 mg	
	ORAL	TABLET	0.15 mg	
POLYETHYLENE GLYCOL 6000 (UNII:30IQX730WE)	BUCCAL	TABLET	70mg	
	ORAL	CAPSULE		3600 mg
	ORAL	CAPSULE, DELAYED RELEASE		135 mg
	ORAL	CAPSULE, EXTENDED RELEASE		302 mg
	ORAL	TABLET		750 mg
	ORAL	TABLET, COATED	106 mg	
	ORAL	TABLET, DELAYED RELEASE		120 mg
	ORAL	TABLET, EXTENDED RELEASE		250mg
	ORAL	TABLET, FILM COATED		132mg
	ORAL	TABLET, FILM COATED, EXTENDED RELEASE	0.5mg	
	ORAL	TABLET, MULTILAYER, EXTENDED RELEASE		10mg
	ORAL	TABLET, ORALLY DISINTEGRATING, DELAYED RELEASE		69mg
	RECTAL	SUPPOSITORY	128mg	
	SUBLINGUAL	TABLET	70mg	
	TOPICAL	CREAM	5%w/w	
	VAGINAL	TABLET	3mg	
	VAGINAL	TABLET, FILM COATED	0.06mg	



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	Functional Category	Regulatory status	Safety
Polyoxylglycerides	Emulsifying agent; modified-release agent; nonionic surfactant; penetration enhancer; solubilizing agent.	Included in the FDA Inactive Ingredients Database (oral capsules and tablets). Included in nonparenteral medicines licensed in the UK. Included in the Substances Added to Food (formerly EAFUS) list compiled by the FDA (PEG fatty acid esters and mono-, di-, and triglycerides mixture). Included in the Canadian Natural Health Products Ingredients Database.	Polyoxylglycerides are used in oral and topical pharmaceutical formulations, and also in cosmetics and food products. They are generally regarded as relatively nonirritant and nontoxic materials.
PEG 6000	Coating agent; ointment base, plasticizing agent, solvent, suppository base, tablet and capsule diluent, tablet and capsule lubricant	Included in the FDA Inactive Ingredients Database (buccal and dental preparations, inhalations, IV, IM, SC and other injectable preparations, nasal and ophthalmic preparations, oral capsules (...)). Included in parenteral and nonparenteral medicines licensed in the UK. Included in the Substances Added to Food (formerly EAFUS) list compiled by the FDA. Accepted for use as a food additive in Europe. Included in the Canadian Natural Health Products Ingredient Database	Polyethylene glycols are widely used in a variety of pharmaceutical formulations. Generally, they are regarded as having a very low order of toxicity and as nonirritant materials. (...) The WHO has set up an estimated acceptable daily intake of PEG at up to 10 mg/kg body-weight.

10.2. Food/Nutraceutical/Dietary supplement applications

Polyoxylglycerides (including Lauroyl Polyoxylglycerides) and polyethylene glycols have a long history of use as food additive for use in nutraceutical products and dietary supplements.

Country	Reference	Registration identifier
E.U.	EFA (European Food additives)	/ E1521: polyethylene glycol
U.S.A.	USFA US Food additives according to 21 CFR	Food Additives Permitted For Direct Addition to Food for Human Consumption: 21 CFR § 172.736 Glycerides and Polyglycides of hydrogenated vegetable oils 21 CFR § 172.820 Polyethylene glycol (mean molecular weight 200-9,500)



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NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s') properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s') suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

